Standard Form & Indices

Quick reference sheet · GCSE Maths, Number · keep handy for revision

1. Laws of indices

Multiplying (same base): $a^m \times a^n = a^{m+n}$ — add the powers.

Dividing (same base): $a^m \div a^n = a^{m-n}$ — subtract the powers.

Power of a power: $(a^m)^n = a^{m \times n}$ — multiply the powers.

These laws only apply when the base is the same on both sides.

e.g. $3^4 \times 3^2 = 3^6$ · $5^7 \div 5^3 = 5^4$ · $(2^3)^2 = 2^6 =$ **64**

2. Zero and negative indices

$\mathbf{a^0 = 1}$ for any non-zero a. e.g. $9^0 = 1$, $100^0 = 1$.

$\mathbf{a^{-n} = 1 \div a^n}$ — a negative index flips the number into a fraction, it does not make the answer negative.

Expression	Value
7^0	1
3^{-1}	1/3
4^{-2}	1/16
10^{-2}	0.01

3. Standard form: writing large and small numbers

Standard form writes any number as $\mathbf{A \times 10^n}$, where $\mathbf{1 \le A < 10}$ and n is an integer. Large numbers (10 or more) use a positive power; small numbers (less than 1) use a negative power.

Ordinary number	Standard form
8,000	8×10^3
450,000	4.5×10^5
0.006	6×10^{-3}
0.00072	7.2×10^{-4}

4. Calculating with standard form

Multiply: $\mathbf{(A \times 10^m) \times (B \times 10^n) = (A \times B) \times 10^{m+n}}$

Divide: $\mathbf{(A \times 10^m) \div (B \times 10^n) = (A \div B) \times 10^{m-n}}$

If the front numbers multiply to 10 or more, adjust back into standard form. e.g. $(5 \times 10^6) \times (4 \times 10^3) = 20 \times 10^9 = \mathbf{2 \times 10^{10}}$.